

# Image Classification Extension

<https://github.com/aiyshwary/Image-Classification-Extension-in-Rapid-Miner>

Python

PyTorch

ONNX

RapidMiner

Transfer Learning

## OVERVIEW

A RapidMiner Studio extension enabling no-code transfer learning for image classification, supporting fine-tuning of popular CNN architectures on custom datasets and ONNX-based inference without writing a single line of code.

## IMPACT

Developed a RapidMiner extension with two custom operators — Fine Tuning and Inference — enabling practitioners to apply transfer learning with ResNet, VGGNet, DenseNet, Inception and more using a drag-and-drop interface.

## TECHNICAL WALKTHROUGH

### 1. Image Classification Extension

#### Problem Statement

In real-world scenarios, image datasets vary significantly in size, quality, and domain. A fixed model often fails to generalize well. Hence, there is a need for a customizable image classification system that can adapt to different datasets while maintaining good accuracy.

#### Approach

Built an image classification extension using TensorFlow and PyTorch

Leveraged transfer learning with pre-trained models such as

1. ResNet

2. AlexNet

3. VGG

4. SqueezeNet

5. DenseNet

6. Inception

7. Vision Transformers (ViT, DeiT via Hugging Face)

8. Fine-Tuning

9. Dataset organized in class-wise folders

10. Automatic 75% training / 25% validation split

### Customizable parameters

1. Number of layers

2. Batch size

3. Learning rate

4. Optimizer

5. Epochs

6. Used data augmentation, class-weighted learning, and callbacks

7. like early stopping and learning-rate scheduling

8. Inference

9. Trained model exported to ONNX

10. Images are preprocessed and passed to the model

### Output includes

1. Predicted class label

2. Confidence score

3. Images are automatically placed into output folders based on

4. predicted class

5. Results

6. Animal classification: ~87% accuracy

7. Facial emotion recognition: ~80% accuracy

8. Flower dataset: ~94% accuracy

#### KEY DETAILS

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1. Fine Tuning operator accepts an image directory and hyperparameters; outputs a fine-tuned model in ONNX format.

2. Inference operator loads the ONNX model and outputs a prediction DataFrame with classified labels.

3. Supports ResNet, AlexNet, VGGNet, SqueezeNet, DenseNet, and Inception architectures.

4. Compatible with both CPU and GPU (CUDA) environments via configurable conda environment.

5. Installed as a JAR extension in RapidMiner Studio with Python Scripting backend.